Mathematics 7 — Solved Exercises. Domain 1, Sub-domain (i) Rational Numbers and Decimal Numbers COMPLETE (book pp. 1-24: Exercises 1.1-1.5, all Skill Practice boxes and the Challenge). Sub-domains (ii)-(vi) and Domains 2-5 are still to be added — this PDF is PARTIAL and is labelled as such.

Class 7 · Punjab Curriculum and Textbook Board (PCTB)

Solved Exercises

PARTIAL — Domain 1 Sub-domain (i) complete only (see title). Sub-domains (ii) Simplification (p. 25), (iii) Sets (p. 29), (iv) Rate, Ratio and Proportion (p. 43), (v) Financial Arithmetic (p. 53), (vi) Squares and Square Roots (p. 66), and Domains 2-5 (pp. 78-218) remain.

Domain 1 — Numbers and Operations: Rational Numbers and Decimal Numbers

Exercise — Section 1.1 (Skill Practice, p. 2)

Q1. Think! Give any five real life examples in which you use negative integers.

Answer: Negative integers are used wherever a value is counted below a zero reference point.

(i) Temperature below 0 °C in winter, for example −5 °C.

(ii) Floors below ground level in a building, for example basement −1 and −2.

(iii) Depth below sea level, for example a fish at −750 ft.

(iv) Money owed or borrowed, for example a debt of Rs. 500 written as −500.

(v) A drop in marks or points, for example losing 10 points written as −10.

Q2. Skill Practice: Solve the following: $|-10|$, $|-17|$, $|19|$ and $|-6|$. Also arrange them in ascending and descending order.

Step 1: The absolute value $|x|$ is the distance of x from 0 on the number line, so it is never negative.

Step 2: $|-10| = 10$, because −10 is 10 units away from 0.

Step 3: $|-17| = 17$, because −17 is 17 units away from 0.

Step 4: $|19| = 19$, because 19 is 19 units away from 0.

Step 5: $|-6| = 6$, because −6 is 6 units away from 0.

Step 6: Ascending order (smallest to largest): 6, 10, 17, 19.

Step 7: Descending order (largest to smallest): 19, 17, 10, 6.

Final answer: $|-10| = 10$, $|-17| = 17$, $|19| = 19$, $|-6| = 6$; ascending 6, 10, 17, 19; descending 19, 17, 10, 6.

Q3. Skill Practice: Represent the following whole numbers on the number line: (i) 880, 730, 540, 990, 670.

Step 1: All five numbers lie between 500 and 1000, so draw a number line from 500 to 1000 with equal gaps of 100.

Step 2: Mark 500, 600, 700, 800, 900 and 1000 at equal distances.

Step 3: 540 lies between 500 and 600, close to 500.

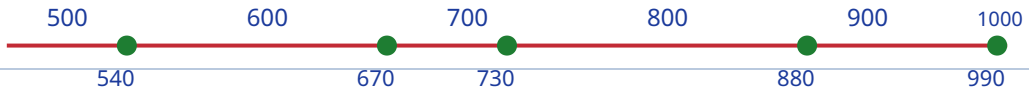
Step 4: 670 lies between 600 and 700, close to 700.

Step 5: 730 lies between 700 and 800, close to 700.

Step 6: 880 lies between 800 and 900, close to 900.

Step 7: 990 lies between 900 and 1000, very close to 1000.

Final answer in ascending order on the line: 540, 670, 730, 880, 990.



Challenge — Section 1.1.2 (book p. 3)

Challenge: Find the 7-digit number with the help of the given clues: 2 is at ones place. The place value of 1 is 100. 7 is not at the tens place. 4 is at thousand place. The digit at the hundred thousands place is sum of digits at thousands and ones

place. The digit at ten thousands is one more than the digit at thousand place. 7 digits are 1, 2, 3, 4, 5, 6 and 7 to make a 7-digit number.

Step 1: Write the seven places from the left: millions, hundred thousands, ten thousands, thousands, hundreds, tens, ones.

Step 2: Ones place = 2 (given).

Step 3: Place value of 1 is 100, so 1 is at the hundreds place.

Step 4: Thousands place = 4 (given).

Step 5: Hundred thousands place = thousands digit + ones digit = 4 + 2 = 6.

Step 6: Ten thousands place = thousands digit + 1 = 4 + 1 = 5.

Step 7: Digits used so far are 2, 1, 4, 6, 5. The digits left from {1, 2, 3, 4, 5, 6, 7} are 3 and 7, for the millions and tens places.

Step 8: 7 is NOT at the tens place, so tens place = 3 and millions place = 7.

Step 9: Fill the places in order: millions 7, hundred thousands 6, ten thousands 5, thousands 4, hundreds 1, tens 3, ones 2.

Final answer: 7,654,132.

Skill Practice — Section 1.1.5 (book p. 4)

Skill Practice: Represent the following on the number line: (i) -700, 100, 900, -800, -500 (ii) 2200, -1800, -900, 1500, 2500.

Step 1: For part (i), draw a number line from -1000 to 1000 with equal gaps of 100.

Step 2: Mark -800 between -900 and -700, -700, -500, 100 and 900 in that left-to-right order.

Step 3: Ascending order (i): -800, -700, -500, 100, 900. Descending order (i): 900, 100, -500, -700, -800.

Step 4: For part (ii), draw a number line from -2000 to 3000 with equal gaps of 100.

Step 5: Mark -1800, -900, 1500, 2200 and 2500 in that left-to-right order.

Final answer (ii): ascending -1800, -900, 1500, 2200, 2500; descending 2500, 2200, 1500, -900, -1800.

Skill Practice: Represent the following integers on number line: -50, 60, -30, -20, 10, 40.

Step 1: Draw a number line from -60 to 70 with equal gaps of 10.

Step 2: Mark the numbers from left to right: -50, -30, -20, 10, 40, 60.

Final answer: ascending order -50, -30, -20, 10, 40, 60; descending order 60, 40, 10, -20, -30, -50.

Skill Practice — Section 1.1.8 (book p. 5)

Skill Practice: Write the following integers on your notebook and arrange them in ascending and descending order: -890.550, -230.430, -410.890.

Step 1: All three numbers are negative, so the one farthest from zero is the smallest.

Step 2: Compare: -890.550 is farthest from zero, then -410.890, then -230.430 is closest to zero.

Final answer: ascending order -890.550, -410.890, -230.430; descending order -230.430, -410.890, -890.550.

Exercise 1.1 — Comparing and Ordering (book p. 5)

Q1 (i). Represent the following on the number line: 230, 130, 150, 100, 170, 120.

Step 1: Draw a number line from 0 to 250, gaps of 10.

Step 2: Mark the numbers from left to right in ascending order: 100, 120, 130, 150, 170, 230.

Final answer: the points sit on the line in the order 100, 120, 130, 150, 170, 230 from left to right.

Q1 (ii). Represent the following on the number line: 70, 80, 95, 35, 40, 45, 60.

Step 1: Draw a number line from 0 to 100, gaps of 5.

Step 2: Mark the numbers from left to right in ascending order: 35, 40, 45, 60, 70, 80, 95.

Final answer: the points sit on the line in the order 35, 40, 45, 60, 70, 80, 95 from left to right.

Q1 (iii). Represent the following on the number line: 1600, 1000, 1700, 1500, 900, 2000.

Step 1: Draw a number line from 0 to 2000, gaps of 100.

Step 2: Mark the numbers from left to right in ascending order: 900, 1000, 1500, 1600, 1700, 2000.

Final answer: the points sit on the line in the order 900, 1000, 1500, 1600, 1700, 2000 from left to right.

Q1 (iv). Represent the following on the number line: -50, 60, -25, -70, -10, -35, 45.

Step 1: Draw a number line from -100 to 100, gaps of 10.

Step 2: Mark the numbers from left to right in ascending order: -70, -50, -35, -25, -10, 45, 60.

Final answer: the points sit on the line in the order -70, -50, -35, -25, -10, 45, 60 from left to right.

Q1 (v). Represent the following on the number line: -225, 440, -400, 100, 150, 120, 110.

Step 1: Draw a number line from -500 to 500, gaps of 50.

Step 2: Mark the numbers from left to right in ascending order: -400, -225, 100, 110, 120, 150, 440.

Final answer: the points sit on the line in the order -400, -225, 100, 110, 120, 150, 440 from left to right.

Q1 (vi). Represent the following on the number line: -1100, 2100, -1500, -2500, 1600.

Step 1: Draw a number line from -3000 to 3000, gaps of 100.

Step 2: Mark the numbers from left to right in ascending order: -2500, -1500, -1100, 1600, 2100.

Final answer: the points sit on the line in the order -2500, -1500, -1100, 1600, 2100 from left to right.

Q1 (vii). Represent the following on the number line: -15.5, 20.7, 11.2, -10.5, -12.5.

Step 1: Draw a number line from -20 to 25, gaps of 1.

Step 2: Mark the numbers from left to right in ascending order: -15.5, -12.5, -10.5, 11.2, 20.7.

Final answer: the points sit on the line in the order -15.5, -12.5, -10.5, 11.2, 20.7 from left to right.

Q1 (viii). Represent the following on the number line: 0.5, -0.7, -0.1, 0.3, 0.9, -0.5.

Step 1: Draw a number line from -1 to 1, gaps of 0.1.

Step 2: Mark the numbers from left to right in ascending order: -0.7, -0.5, -0.1, 0.3, 0.5, 0.9.

Final answer: the points sit on the line in the order -0.7, -0.5, -0.1, 0.3, 0.5, 0.9 from left to right.

Q2 (i). Compare by using symbol > or <. Also arrange them in ascending and descending order: 235691, 235090, 245091, 245192.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 245192 > 245091 > 235691 > 235090.

Final answer: 245192 > 245091 > 235691 > 235090; ascending order: 235090, 235691, 245091, 245192.

Q2 (ii). Compare by using symbol > or <. Also arrange them in ascending and descending order: 578901, 579803, 679807, 679817.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 679817 > 679807 > 579803 > 578901.

Final answer: 679817 > 679807 > 579803 > 578901; ascending order: 578901, 579803, 679807, 679817.

Q2 (iii). Compare by using symbol > or <. Also arrange them in ascending and descending order: 10028, 100026, 110028, 110027.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 110028 > 110027 > 100026 > 10028.

Final answer: 110028 > 110027 > 100026 > 10028; ascending order: 10028, 100026, 110027, 110028.

Q2 (iv). Compare by using symbol > or <. Also arrange them in ascending and descending order: 562389, 562399, 572390, 572381.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 572390 > 572381 > 562399 > 562389.

Final answer: 572390 > 572381 > 562399 > 562389; ascending order: 562389, 562399, 572381, 572390.

Q3 (i). Compare by using symbol > or <. Also arrange them in ascending and descending order: -8395, -8496, -8491, -8394.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: -8394 > -8395 > -8491 > -8496.

Final answer: -8394 > -8395 > -8491 > -8496; ascending order: -8496, -8491, -8395, -8394.

Q3 (ii). Compare by using symbol > or <. Also arrange them in ascending and descending order: 503, 530, 556, -563.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 556 > 530 > 503 > -563.

Final answer: 556 > 530 > 503 > -563; ascending order: -563, 503, 530, 556.

Q3 (iii). Compare by using symbol > or <. Also arrange them in ascending and descending order: -137, -138, -1886, 1308.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 1308 > -137 > -138 > -1886.

Final answer: 1308 > -137 > -138 > -1886; ascending order: -1886, -138, -137, 1308.

Q3 (iv). Compare by using symbol > or <. Also arrange them in ascending and descending order: -87650, -78432, -78402, 78401.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 78401 > -78402 > -78432 > -87650.

Final answer: 78401 > -78402 > -78432 > -87650; ascending order: -87650, -78432, -78402, 78401.

Q4 (i). Compare by using symbol > or <. Also arrange them in ascending and descending order: 28.356, 28.343, 28.357, 28.532.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 28.532 > 28.357 > 28.356 > 28.343.

Final answer: 28.532 > 28.357 > 28.356 > 28.343; ascending order: 28.343, 28.356, 28.357, 28.532.

Q4 (ii). Compare by using symbol > or <. Also arrange them in ascending and descending order: 120.08, 130.08, 120.8, 120.01.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 130.08 > 120.8 > 120.08 > 120.01.

Final answer: 130.08 > 120.8 > 120.08 > 120.01; ascending order: 120.01, 120.08, 120.8, 130.08.

Q4 (iii). Compare by using symbol > or <. Also arrange them in ascending and descending order: 131.01, 131.08, 103.78, 113.08.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 131.08 > 131.01 > 113.08 > 103.78.

Final answer: 131.08 > 131.01 > 113.08 > 103.78; ascending order: 103.78, 113.08, 131.01, 131.08.

Q4 (iv). Compare by using symbol > or <. Also arrange them in ascending and descending order: 236.089, 236.219, 236.217, 236.207.

Step 1: Compare the numbers two at a time from the greatest place value.

Step 2: In descending order they are: 236.219 > 236.217 > 236.207 > 236.089.

Final answer: 236.219 > 236.217 > 236.207 > 236.089; ascending order: 236.089, 236.207, 236.217, 236.219.

Exercise 1.2 — Rational Numbers (book pp. 10-11)

Q1 (i). Represent the following rational number on number line: 1/3.

Step 1: 1/3 = 1/3 in value.

Step 2: On the number line it lies between 0 and 1.

Final answer: mark 1/3 between 0 and 1 on the number line.

Q1 (ii). Represent the following rational number on number line: 3/7.

Step 1: 3/7 = 3/7 in value.

Step 2: On the number line it lies between 0 and 1.

Final answer: mark 3/7 between 0 and 1 on the number line.

Q1 (iii). Represent the following rational number on number line: -3/5.

Step 1: -3/5 = -3/5 in value.

Step 2: On the number line it lies between -1 and 0.

Final answer: mark -3/5 between -1 and 0 on the number line.

Q1 (iv). Represent the following rational number on number line: 5/3.

Step 1: 5/3 = 1 2/3 in value.

Step 2: On the number line it lies between 1 and 2.

Final answer: mark 5/3 between 1 and 2 on the number line.

Q1 (v). Represent the following rational number on number line: 5/8.

Step 1: 5/8 = 5/8 in value.

Step 2: On the number line it lies between 0 and 1.

Final answer: mark 5/8 between 0 and 1 on the number line.

Q1 (vi). Represent the following rational number on number line: 3/10.

Step 1: 3/10 = 3/10 in value.

Step 2: On the number line it lies between 0 and 1.

Final answer: mark 3/10 between 0 and 1 on the number line.

Q1 (vii). Represent the following rational number on number line: -2 1/2.

Step 1: -2 1/2 = -2 1/2 in value.

Step 2: On the number line it lies between -3 and -2.

Final answer: mark -2 1/2 between -3 and -2 on the number line.

Q1 (viii). Represent the following rational number on number line: -2 3/4.

Step 1: -2 3/4 = -2 3/4 in value.

Step 2: On the number line it lies between -3 and -2.

Final answer: mark -2 3/4 between -3 and -2 on the number line.

Q2. Convert: (i) 1 2/3, 3 4/5, 2 5/10, 8 3/7 into improper fractions (ii) 18/7, 63/14,

43/13, 115/9 into mixed fractions.

Step 1: Mixed to improper: multiply the whole number by the denominator and add the numerator.

Step 2: (i) 1 2/3 = (1 x 3 + 2)/3 = 5/3; 3 4/5 = (3 x 5 + 4)/5 = 19/5; 2 5/10 = (2 x 10 + 5)/10 = 25/10 = 5/2; 8 3/7 = (8 x 7 + 3)/7 = 59/7.

Step 3: Improper to mixed: divide the numerator by the denominator; the quotient is the whole part and the remainder over the denominator is the fraction.

Step 4: (ii) 18/7 = 2 4/7 (18 = 2 x 7 + 4); 63/14 = 4 7/14 = 4 1/2; 43/13 = 3 4/13 (43 = 3 x 13 + 4); 115/9 = 12 7/9 (115 = 12 x 9 + 7).

Final answer: (i) 5/3, 19/5, 25/10, 59/7 (ii) 2 4/7, 4 1/2, 3 4/13, 12 7/9.

Q3 (i). Put the correct sign (>, < or =) between the following rational numbers: 3/5 and 6/25.

Step 1: Compare the two numbers by value.

Step 2: 3/5 > 6/25.

Final answer: 3/5 > 6/25.

Q3 (ii). Put the correct sign (>, < or =) between the following rational numbers: 7/10 and -7/10.

Step 1: Compare the two numbers by value.

Step 2: 7/10 > -7/10.

Final answer: 7/10 > -7/10.

Q3 (iii). Put the correct sign (>, < or =) between the following rational numbers:

-3/8 and -5/8.

Step 1: Compare the two numbers by value.

Step 2: -3/8 > -5/8.

Final answer: -3/8 > -5/8.

Q3 (iv). Put the correct sign (>, < or =) between the following rational numbers: 1

1/7 and 2 3/14.

Step 1: Compare the two numbers by value.

Step 2: 1 1/7 < 2 3/14.

Final answer: 1 1/7 < 2 3/14.

Q3 (v). Put the correct sign (>, < or =) between the following rational numbers: 1

5/12 and 7/12.

Step 1: Compare the two numbers by value.

Step 2: 1 5/12 > 7/12.

Final answer: 1 5/12 > 7/12.

Q3 (vi). Put the correct sign (>, < or =) between the following rational numbers:

-19/25 and -7/25.

Step 1: Compare the two numbers by value.

Step 2: -19/25 < -7/25.

Final answer: -19/25 < -7/25.

Q4 (i). Compare the following rational numbers: 2/7 , 2/5.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: 2/7 < 2/5.

Q4 (ii). Compare the following rational numbers: 5/9 , -3/5.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: 5/9 > -3/5.

Q4 (iii). Compare the following rational numbers: 1/12 , -1/12.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: 1/12 > -1/12.

Q4 (iv). Compare the following rational numbers: -2/7 , -3/11.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: -2/7 < -3/11.

Q4 (v). Compare the following rational numbers: -5/11 , -7/9.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: -5/11 > -7/9.

Q4 (vi). Compare the following rational numbers: 3/22 , -5/22.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: 3/22 > -5/22.

Q4 (vii). Compare the following rational numbers: 3 5/9 , 4 5/18.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: 3 5/9 < 4 5/18.

Q4 (viii). Compare the following rational numbers: -1 7/25 , -3/5.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: -1 7/25 < -3/5.

Q4 (ix). Compare the following rational numbers: 3 1/2 , 2 1/14.

Step 1: Every positive rational number is greater than every negative rational number.

Step 2: For numbers of the same sign, compare by making the denominators equal (LCM) or by cross-multiplication.

Final answer: 3 1/2 > 2 1/14.

Q5 (i). Arrange the following rational numbers in ascending order: 1/4, 3/5, 7/8, 1 1/4.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators (for negative numbers, the smaller numerator value is the smaller number).

Final answer: ascending order 1/4, 3/5, 7/8, 1 1/4.

Q5 (ii). Arrange the following rational numbers in ascending order: 6/7, 3/14, -1 2/7, -13/14.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators (for negative numbers, the smaller numerator value is the smaller number).

Final answer: ascending order -1 2/7, -13/14, 3/14, 6/7.

Q5 (iii). Arrange the following rational numbers in ascending order: -7/12, -1 1/4, -2/3, 5/8.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators (for negative numbers, the smaller numerator value is the smaller number).

Final answer: ascending order -1 1/4, -2/3, -7/12, 5/8.

Q5 (iv). Arrange the following rational numbers in ascending order: 7/8, -1 1/4, -1 2/5, 1 5/8.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators (for negative numbers, the smaller numerator value is the smaller number).

Final answer: ascending order -1 2/5, -1 1/4, 7/8, 1 5/8.

Q6 (i). Arrange the following rational numbers in descending order: 2/3, 4/7, -4/-7, -3/4.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators and write the numbers from the largest to the smallest. Note: -4/-7 = 4/7, so 4/7 appears twice in this list, as printed in the book.

Final answer: descending order 2/3, 4/7, 4/7, -3/4.

Q6 (ii). Arrange the following rational numbers in descending order: 4/5, 13/30, 7/10, -7/15.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators and write the numbers from the largest to the smallest.

Final answer: descending order 4/5, 7/10, 13/30, -7/15.

Q6 (iii). Arrange the following rational numbers in descending order: 5/8, 13/24, -1 5/12, 3/16.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators and write the numbers from the largest to the smallest.

Final answer: descending order 5/8, 13/24, 3/16, -1 5/12.

Q6 (iv). Arrange the following rational numbers in descending order: -5/11, -9/22, -4/33, -41/44.

Step 1: Make all denominators equal by taking the LCM of the denominators.

Step 2: Compare the numerators and write the numbers from the largest to the smallest.

Final answer: descending order -4/33, -9/22, -5/11, -41/44.

Skill Practice — Section 1.1.14 (book pp. 11-14)

Skill Practice: Solve the following: 1/2 + 7/9 + 1/9 = ?

Step 1: Add the fractions with the same denominator first: 7/9 + 1/9 = 8/9.

Step 2: Now add 1/2 + 8/9. The LCM of 2 and 9 is 18.

Step 3: 1/2 = 9/18 and 8/9 = 16/18.

Step 4: 9/18 + 16/18 = 25/18.

Final answer: 25/18 = 1 7/18.

Skill Practice: Danish purchased a cake. He eats 3/10 parts of the cake. Danish's friend eats 1/5 parts of the cake and his sister eats 4/15 parts of the cake. Can you find out how much cake is left with Danish?

Step 1: Total eaten = 3/10 + 1/5 + 4/15. The LCM of 10, 5 and 15 is 30.

Step 2: 3/10 = 9/30, 1/5 = 6/30, 4/15 = 8/30.

Step 3: 9/30 + 6/30 + 8/30 = 23/30.

Step 4: Cake left = 1 - 23/30 = 7/30.

Final answer: 7/30 of the cake is left.

Skill Practice: Solve the following: 11/15 - 9/15 - 2/15 = ?

Step 1: All three fractions have the same denominator 15, so subtract the numerators.

Step 2: 11 - 9 - 2 = 0.

Step 3: 0/15 = 0.

Final answer: 0.

Skill Practice: Find the additive inverse: (i) 2/3 (ii) -4/7.

Step 1: The additive inverse of a number is the same number with the opposite sign.

Step 2: (i) The additive inverse of 2/3 is -2/3, because 2/3 + (-2/3) = 0.

Step 3: (ii) The additive inverse of -4/7 is 4/7, because -4/7 + 4/7 = 0.

Final answer: (i) -2/3 (ii) 4/7.

Skill Practice: Ahmad's friend left 3/8 of a whole pizza for him, Ahmad ate 1/3 of what was left. How much of the whole pizza did Ahmad left?

Step 1: What was left for Ahmad = 3/8 of the pizza.

Step 2: Ahmad ate 1/3 of what was left: 1/3 x 3/8 = 3/24 = 1/8 of the whole pizza.

Step 3: What Ahmad left = 3/8 - 1/8 = 2/8.

Step 4: 2/8 = 1/4.

Final answer: 1/4 of the whole pizza was left.

Skill Practice: Solve the following: (2/3 x 9/10) / (3/10) = ?

Step 1: First multiply inside the brackets: 2/3 x 9/10 = 18/30 = 3/5.

Step 2: Now divide by 3/10: 3/5 / (3/10) = 3/5 x 10/3.

Step 3: 3/5 x 10/3 = 30/15 = 2.

Final answer: 2.

Exercise 1.3 — Operations on Rational Numbers (book pp. 15-16)

Q1 (i). Add the following rational numbers: 5/9 and 4/9.

Step 1: LCM = 9. 5/9 = 5/9 = 4/9 = 4/9. Sum of numerators over 9 = 1.

Final answer: 1.

Q1 (ii). Add the following rational numbers: -3 1/4 and 9/14.

Step 1: LCM = 28. -3 1/4 = -91/28 = 9/14 = 18/28. Sum of numerators over 28 = -2

17/28.

Final answer: -2 17/28.

Q1 (iii). Add the following rational numbers: -7/11 and -3/22.

Step 1: LCM = 22. -7/11 = -14/22 = -3/22 = -3/22. Sum of numerators over 22 =

-17/22.

Final answer: -17/22.

Q1 (iv). Add the following rational numbers: -6/13 and -8/13.

Step 1: LCM = 13. -6/13 = -6/13 = -8/13 = -8/13. Sum of numerators over 13 = -1

1/13.

Final answer: -1 1/13.

Q1 (v). Add the following rational numbers: 3/4 and -1 1/5.

Step 1: LCM = 20. 3/4 = 15/20 = -1 1/5 = -24/20. Sum of numerators over 20 =

-9/20.

Final answer: -9/20.

Q1 (vi). Add the following rational numbers: 4 and -3/5.

Step 1: LCM = 5. 4 = 20/5 = -3/5 = -3/5. Sum of numerators over 5 = 3 2/5.

Final answer: 3 2/5.

Q2 (i). Solve the following: -3/40 + -5/14.

Step 1: LCM = 280. -3/40 = -21/280 = -5/14 = -100/280. Sum of numerators over 280 = -121/280.

Final answer: -121/280.

Q2 (ii). Solve the following: -5/9 + -1/3.

Step 1: LCM = 9. -5/9 = -5/9 = -1/3 = -3/9. Sum of numerators over 9 = -8/9.

Final answer: -8/9.

Q2 (iii). Solve the following: 3/4 + -2/5 + -7/20.

Step 1: LCM = 20. 3/4 = 15/20 = -2/5 = -8/20 = -7/20 = -7/20. Sum of numerators over 20 = 0.

Final answer: 0.

Q2 (iv). Solve the following: -2 + 7/11 + -4/15.

Step 1: LCM = 165. -2 = -330/165 = 7/11 = 105/165 = -4/15 = -44/165. Sum of numerators over 165 = -1 104/165.

Final answer: -1 104/165.

Q2 (v). Solve the following: -1/8 + 5/12 + -7/24.

Step 1: LCM = 24. -1/8 = -3/24 = 5/12 = 10/24 = -7/24 = -7/24. Sum of numerators over 24 = 0.

Final answer: 0.

Q3 (i). Subtract: 3/4 from 1/2.

Step 1: 1/2 - 3/4 = 1/2 + (-3/4).

Step 2: LCM = 4. 1/2 = 2/4 = -3/4 = -3/4. Sum of numerators over 4 = -1/4.

Final answer: -1/4.

Q3 (ii). Subtract: -1 1/6 from 5/18.

Step 1: 5/18 - -1 1/6 = 5/18 + (1 1/6).

Step 2: LCM = 18. 5/18 = 5/18 = 1 1/6 = 21/18. Sum of numerators over 18 = 1 4/9.

Final answer: 1 4/9.

Q3 (iii). Subtract: -4/9 from -3/5.

Step 1: -3/5 - -4/9 = -3/5 + (4/9).

Step 2: LCM = 45. -3/5 = -27/45 = 4/9 = 20/45. Sum of numerators over 45 = -7/45.

Final answer: -7/45.

Q3 (iv). Subtract: -7/8 from -2.

Step 1: -2 - -7/8 = -2 + (7/8).

Step 2: LCM = 8. -2 = -16/8 = 7/8 = 7/8. Sum of numerators over 8 = -1 1/8.

Final answer: -1 1/8.

Q4 (i). Solve the following: 4/7 - 5/14.

Step 1: Write both numbers with a common denominator (LCM of the

denominators).

Step 2: Subtract the numerators.

Final answer: 3/14.

Q4 (ii). Solve the following: 1 1/5 - 1/2.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: 7/10.

Q4 (iii). Solve the following: -4/15 - 8/25.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: -44/75.

Q4 (iv). Solve the following: -2/5 - 3/14.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: -43/70.

Q4 (v). Solve the following: 4/9 - 5/9.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: -1/9.

Q4 (vi). Solve the following: 1 1/12 - -4/9.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: 1 19/36.

Q4 (vii). Solve the following: 5/8 - -3/7.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: 1 3/56.

Q4 (viii). Solve the following: 4/11 - -5/22.

Step 1: Write both numbers with a common denominator (LCM of the denominators).

Step 2: Subtract the numerators.

Final answer: 13/22.

Q5 (i). Solve the following: 7/11 x 5/6.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: 35/66.

Q5 (ii). Solve the following: 3/7 x -4/5.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: -12/35.

Q5 (iii). Solve the following: -2/9 x 5/12.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: -5/54.

Q5 (iv). Solve the following: -3/14 x -5/7.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: 15/98.

Q5 (v). Solve the following: -1 1/6 x -18.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: 21.

Q5 (vi). Solve the following: -2 3/5 x -25/26.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: 2 1/2.

Q5 (vii). Solve the following: -5/11 x -1 7/22.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: 145/242.

Q5 (viii). Solve the following: -4 1/2 x -1 3/4.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: 7 7/8.

Q5 (ix). Solve the following: -5/8 x 48.

Step 1: Multiply the numerators together and the denominators together.

Step 2: Cancel common factors and write the result in its simplest form.

Final answer: -30.

Q6 (i). Solve the following: 5/6 / 7/12.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: 5/6 x 1 5/7 = 1 3/7.

Final answer: 1 3/7.

Q6 (ii). Solve the following: -7/16 / 1/8.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: -7/16 x 8 = -3 1/2.

Final answer: -3 1/2.

Q6 (iii). Solve the following: -1 1/5 / 8.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: -1 1/5 x 1/8 = -3/20.

Final answer: -3/20.

Q6 (iv). Solve the following: 1 1/15 / -18.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: 1 1/15 x -1/18 = -8/135.

Final answer: -8/135.

Q6 (v). Solve the following: -3/10 / -21/40.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: -3/10 x -1 19/21 = 4/7.

Final answer: 4/7.

Q6 (vi). Solve the following: 4 1/16 / -1 5/8.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: 4 1/16 x -8/13 = -2 1/2.

Final answer: -2 1/2.

Q6 (vii). Solve the following: -12 / -5/6.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: -12 x -1 1/5 = 14 2/5.

Final answer: 14 2/5.

Q6 (viii). Solve the following: -1 10/19 / -5/58.

Step 1: Division by a rational number is multiplication by its reciprocal.

Step 2: -1 10/19 x -11 3/5 = 17 67/95.

Final answer: 17 67/95.

Q7. The product of two rational numbers is -1/10. If one number is -3/5, find the other.

Step 1: Other number = product / known number = (-1/10) / (-3/5).

Step 2: (-1/10) x (-5/3) = 5/30.

Step 3: 5/30 = 1/6.

Final answer: 1/6.

Q8. By what rational number should we multiply -15/14 to get -3/7?

Step 1: Required number = (-3/7) / (-15/14).

Step 2: (-3/7) x (-14/15) = 42/105.

Step 3: 42/105 = 2/5.

Final answer: 2/5.

Q9. Find the additive inverse of: (i) 3/5 (ii) -7/12 (iii) -(-8/15) (iv) -7 5/12 (v) -2 1/13.

Step 1: The additive inverse changes the sign of the number.

Step 2: (i) -3/5 (ii) 7/12 (iii) -(-8/15) = 8/15, so its inverse is -8/15.

Step 3: (iv) 7 5/12 (v) 2 1/13.

Final answer: (i) -3/5 (ii) 7/12 (iii) -8/15 (iv) 7 5/12 (v) 2 1/13.

Q10. Amna purchased 25/3 m cloth to stitch a shirt. If she used 13/9 m cloth to stitch the shirt. How much cloth was left with her?

Step 1: Cloth left = 25/3 - 13/9. The LCM of 3 and 9 is 9.

Step 2: 25/3 = 75/9.

Step 3: 75/9 - 13/9 = 62/9.

Final answer: 62/9 m = 6 8/9 m of cloth was left.

Q11. If 75/4 kilogram rice is to be packed in 25/8 kilogram packets, then find: (a) How many packets of rice will be packed? (b) How much rice will be there in 20 packets of mass 24/15 kilograms each?

Step 1: (a) Number of packets = (75/4) / (25/8) = 75/4 x 8/25 = 600/100 = 6.

Step 2: (b) Rice in 20 packets = 20 x 24/15 = 480/15 = 32.

Final answer: (a) 6 packets (b) 32 kilograms.

Q12. Zara bought 235/7 litres cooking oil. She used 2 2/14 litres oil for cooking. How much oil was left with her?

Step 1: Oil used = 2 2/14 litres = 30/14 = 15/7 litres.

Step 2: Oil left = 235/7 - 15/7 = 220/7.

Step 3: 220/7 = 31 3/7.

Final answer: 31 3/7 litres of oil was left.

Exercise 1.4 — Properties of Rational Numbers (book pp. 19-20)

Q1 (i). Verify the following commutative property: -3/5 + 6/7 = 6/7 + -3/5.

Step 1: Left hand side = 9/35.

Step 2: Right hand side = 9/35.

Step 3: Both sides are equal.

Final answer: verified — L.H.S. = R.H.S. = 9/35.

Q1 (ii). Verify the following commutative property: 7/10 + -2/5 = -2/5 + 7/10.

Step 1: Left hand side = 3/10.

Step 2: Right hand side = 3/10.

Step 3: Both sides are equal.

Final answer: verified — L.H.S. = R.H.S. = 3/10.

Q1 (iii). Verify the following commutative property: -1 1/2 x -5/7 = -5/7 x -1 1/2.

Step 1: Left hand side = 1 1/14.

Step 2: Right hand side = 1 1/14.

Step 3: Both sides are equal.

Final answer: verified — L.H.S. = R.H.S. = 1 1/14.

Q1 (iv). Verify the following commutative property: 15/26 x 1/3 = 1/3 x 15/26.

Step 1: Left hand side = 5/26.

Step 2: Right hand side = 5/26.

Step 3: Both sides are equal.

Final answer: verified — L.H.S. = R.H.S. = 5/26.

Q2 (i). Verify the following associative property: (1 1/3 + 2/5) + 3/7 = 1 1/3 + (2/5 + 3/7).

Step 1: Left hand side: (1 1/3 + 2/5) + 3/7 = 1 11/15 + 3/7 = 2 17/105.

Step 2: Right hand side: 1 1/3 + (2/5 + 3/7) = 1 1/3 + 29/35 = 2 17/105.

Final answer: verified — L.H.S. = R.H.S. = 2 17/105.

Q2 (ii). Verify the following associative property: (5/6 + -5 1/2) + -3/4 = 5/6 + (-5 1/2 + -3/4).

Step 1: Left hand side: (5/6 + -5 1/2) + -3/4 = -4 2/3 + -3/4 = -5 5/12.

Step 2: Right hand side: 5/6 + (-5 1/2 + -3/4) = 5/6 + -6 1/4 = -5 5/12.

Final answer: verified — L.H.S. = R.H.S. = -5 5/12.

Q2 (iii). Verify the following associative property: (3/4 + -1/2) + 3/10 = 3/4 + (-1/2 + 3/10).

Step 1: Left hand side: (3/4 + -1/2) + 3/10 = 1/4 + 3/10 = 11/20.

Step 2: Right hand side: 3/4 + (-1/2 + 3/10) = 3/4 + -1/5 = 11/20.

Final answer: verified — L.H.S. = R.H.S. = 11/20.

Q2 (iv). Verify the following associative property: (-1/2 + -1/2) + 1/2 = -1/2 + (-1/2 + 1/2).

Step 1: Left hand side: (-1/2 + -1/2) + 1/2 = -1 + 1/2 = -1/2.

Step 2: Right hand side: -1/2 + (-1/2 + 1/2) = -1/2 + 0 = -1/2.

Final answer: verified — L.H.S. = R.H.S. = -1/2.

Q3 (i). Verify the following distributive property: -2/5 x (1/3 + 1 1/3) = (-2/5 x 1/3) + (-2/5 x 1 1/3).

Step 1: Left hand side = -2/3.

Step 2: Right hand side = -2/3.

Final answer: verified — L.H.S. = R.H.S. = -2/3.

Q3 (ii). Verify the following distributive property: 10/11 x (6/7 + 2 1/4) = (10/11 x 6/7) + (10/11 x 2 1/4).

Step 1: Left hand side = 2 127/154.

Step 2: Right hand side = 2 127/154.

Final answer: verified — L.H.S. = R.H.S. = 2 127/154.

Q3 (iii). Verify the following distributive property: 15/16 x (2/3 - 1/3) = (15/16 x 2/3) - (15/16 x 1/3).

Step 1: Left hand side = 5/16.

Step 2: Right hand side = 5/16.

Final answer: verified — L.H.S. = R.H.S. = 5/16.

Q3 (iv). Verify the following distributive property: -2/3 x (4/5 - 4/5) = (-2/3 x 4/5) - (-2/3 x 4/5).

Step 1: Left hand side = 0.

Step 2: Right hand side = 0.

Final answer: verified — L.H.S. = R.H.S. = 0.

Skill Practice — Rounding (book pp. 20-22)

Skill Practice: If the price of a packet of chips is Rs. 15.15, tell the price of packet of chips after rounding.

Step 1: Rounding to the nearest rupee, look at the first decimal digit: 1.

Step 2: 1 is less than 5, so the rupee figure stays 15.

Final answer: Rs. 15.

Skill Practice: Round 48163 to the given degree of accuracy: (i) 2 significant figures (ii) 3 significant figures.

Step 1: (i) To 2 significant figures: the first two digits are 4 and 8, the next digit is 1, which is less than 5, so 48 stays unchanged.

Step 2: Fill the remaining places with zeros: 48000.

Step 3: (ii) To 3 significant figures: the first three digits are 4, 8 and 1, the next digit is 6, which is 5 or more, so 481 rounds up to 482.

Final answer: (i) 48000 (ii) 48200.

Skill Practice: Hamza purchased two litres of petrol for Rs. 438.75. Can you round the number to the given degree of accuracy? (i) 3 significant figures (ii) 1 decimal place.

Step 1: (i) To 3 significant figures: digits 4, 3, 8, next digit 7 is 5 or more, so 438 rounds up to 439.

Step 2: (ii) To 1 decimal place: look at the second decimal, 5, which is 5 or more, so 438.7 rounds up to 438.8.

Final answer: (i) Rs. 439 (ii) Rs. 438.8.

Skill Practice: How many significant figures does 0.0789 have?

Step 1: In a decimal number, zeros before the first non-zero digit are not significant.

Step 2: In 0.0789 the significant digits are 7, 8 and 9.

Final answer: 3 significant figures.

Skill Practice: Round -53278 to the 2 significant figures and 3 significant figures.

Step 1: Round the magnitude 53278. To 2 significant figures: digits 5 and 3, next digit 2 is less than 5, so it stays 53, giving 53000.

Step 2: To 3 significant figures: digits 5, 3 and 2, next digit 7 is 5 or more, so 532 rounds up to 533, giving 53300.

Step 3: Put the minus sign back.

Final answer: (i) -53000 (ii) -53300.

Skill Practice: Round 72/33 to the: (i) 3 decimal places (ii) 2 significant figures (iii) 3 significant figures.

Step 1: 72/33 = 2.181818...

Step 2: (i) To 3 decimal places: look at the fourth decimal, 8, which is 5 or more, so 2.181 rounds up to 2.182.

Step 3: (ii) To 2 significant figures: digits 2 and 1, next digit 8 is 5 or more, so 21 rounds up to 22, giving 2.2.

Step 4: (iii) To 3 significant figures: digits 2, 1 and 8, next digit 1 is less than 5, so it stays 2.18.

Final answer: (i) 2.182 (ii) 2.2 (iii) 2.18.

Skill Practice: 28.3 x 15.2. Find the approximated value and error. Also explain regarding the reasonableness of the solution.

Step 1: Approximated value: round the inputs to whole numbers: 28.3 becomes 28 and 15.2 becomes 15.

Step 2: Approximated product = 28 x 15 = 420.

Step 3: Accurate value = 28.3 x 15.2 = 430.16.

Step 4: Error = |430.16 - 420| = 10.16.

Final answer: approximated 420, accurate 430.16, error 10.16 — the estimate is reasonable because it is close to the accurate value (the error is only about 2% of the product).

Exercise 1.5 — Rounding (book p. 23)

Q1. Round the following to the given degree of accuracy: (i) 87932 (2 significant figures) (ii) 3890 (3 significant figures) (iii) 790253 (2 significant figures) (iv) 10025 (3 significant figures) (v) 18954 (3 significant figures) (vi) 25980 (3 significant figures).

Rule: keep the required significant figures; look at the next digit — if it is 5 or more, round the last kept digit up; if less than 5, leave it unchanged; fill the remaining places with zeros.

(i) 87932 to 2 s.f.: first two digits are 8 and 7, next digit is 9, which is 5 or more, so 87 rounds up to 88.

Final answer (i): 88000.

(ii) 3890 to 3 s.f.: first three digits are 3, 8 and 9, next digit is 0, which is less than 5, so 389 stays unchanged.

Final answer (ii): 3890.

(iii) 790253 to 2 s.f.: first two digits are 7 and 9, next digit is 0, which is less than 5, so 79 stays unchanged.

Final answer (iii): 790000.

(iv) 10025 to 3 s.f.: first three digits are 1, 0 and 0, next digit is 2, which is less than 5, so 100 stays unchanged.

Final answer (iv): 10000.

(v) 18954 to 3 s.f.: first three digits are 1, 8 and 9, next digit is 5, which is 5 or more, so 189 rounds up to 190.

Final answer (v): 19000.

(vi) 25980 to 3 s.f.: first three digits are 2, 5 and 9, next digit is 8, which is 5 or more, so 259 rounds up to 260.

Final answer (vi): 26000.

Q2. Round the numbers away from zero (0) as given: (i) -27.3289 (3 significant figures) (ii) -59058 (2 significant figures) (iii) -12569 (3 significant figures) (iv) 17.238 (2 significant figures) (v) 0.003857 (3 significant figures) (vi) 0.0467453 (3 significant figures).

Rule: round the magnitude (ignore the minus sign) to the required significant figures, then put the minus sign back. 'Away from zero' only changes the result when the rounding digit is exactly 5 followed by nothing; none of these parts is an exact tie.

(i) |-27.3289| = 27.3289 to 3 s.f.: digits 2, 7, 3, next digit 2 is less than 5, so it stays 27.3.

Final answer (i): -27.3.

(ii) |-59058| = 59058 to 2 s.f.: digits 5, 9, next digit 0 is less than 5, so it stays 59, followed by zeros.

Final answer (ii): -59000.

(iii) |-12569| = 12569 to 3 s.f.: digits 1, 2, 5, next digit 6 is 5 or more, so 125 rounds up to 126.

Final answer (iii): -12600.

(iv) 17.238 to 2 s.f.: digits 1, 7, next digit 2 is less than 5, so it stays 17.

Final answer (iv): 17.

(v) 0.003857 to 3 s.f.: the significant digits start at 3, so digits are 3, 8, 5, next digit 7 is 5 or more, so 385 rounds up to 386.

Final answer (v): 0.00386.

(vi) 0.0467453 to 3 s.f.: the significant digits start at 4, so digits are 4, 6, 7, next digit 4 is less than 5, so 467 stays unchanged.

Final answer (vi): 0.0467.

Q3. Round the following to the stated number of decimal places or significant figures: (i) 47/8 (2 decimal places) (ii) 5/27 (3 significant figures) (iii) 22/27 (3 decimal places) (iv) 27/19 (3 significant figures) (v) 37/53 (3 decimal places) (vi) 77/83 (3 significant figures).

Step 1: First divide each fraction to get its decimal value, then round as stated.

(i) 47/8 = 5.875. To 2 decimal places: look at the third decimal, 5, which is 5 or more, so 5.87 rounds up to 5.88.

Final answer (i): 5.88.

(ii) 5/27 = 0.185185... To 3 s.f.: digits are 1, 8, 5, next digit 1 is less than 5, so it stays 0.185.

Final answer (ii): 0.185.

(iii) 22/27 = 0.814814... To 3 decimal places: look at the fourth decimal, 8, which is 5 or more, so 0.814 rounds up to 0.815.

Final answer (iii): 0.815.

(iv) 27/19 = 1.421052... To 3 s.f.: digits are 1, 4, 2, next digit 1 is less than 5, so it stays 1.42.

Final answer (iv): 1.42.

(v) 37/53 = 0.698113... To 3 decimal places: look at the fourth decimal, 1, which is less than 5, so it stays 0.698.

Final answer (v): 0.698.

(vi) 77/83 = 0.927710... To 3 s.f.: digits are 9, 2, 7, next digit 7 is 5 or more, so 927 rounds up to 928.

Final answer (vi): 0.928.

Q4. Round the following to the required degree of accuracy: (i) 25.359 (2 decimal places) (ii) 0.002897 (3 significant figures) (iii) 0.0878 (3 decimal places) (iv)

17.00597 (3 significant figures) (v) 0.0005246 (3 significant figures) (vi) 13.5876 (3 decimal places).

(i) 25.359 to 2 d.p.: look at the third decimal, 9, which is 5 or more, so 25.35 rounds up to 25.36.

Final answer (i): 25.36.

(ii) 0.002897 to 3 s.f.: significant digits are 2, 8, 9, next digit 7 is 5 or more, so 289 rounds up to 290.

Final answer (ii): 0.00290 (the final zero is kept to show three significant figures).

(iii) 0.0878 to 3 d.p.: look at the fourth decimal, 8, which is 5 or more, so 0.087 rounds up to 0.088.

Final answer (iii): 0.088.

(iv) 17.00597 to 3 s.f.: significant digits are 1, 7, 0, next digit 0 is less than 5, so it stays 17.0.

Final answer (iv): 17.0 (the zero after the point is kept to show three significant figures).

(v) 0.0005246 to 3 s.f.: significant digits are 5, 2, 4, next digit 6 is 5 or more, so 524 rounds up to 525.

Final answer (v): 0.000525.

(vi) 13.5876 to 3 d.p.: look at the fourth decimal, 6, which is 5 or more, so 13.587 rounds up to 13.588.

Final answer (vi): 13.588.

Q5. Compare your approximated value and accurate value. Also check is the solution reasonable or not? (i) 4.281 / 2.157 (2 decimal places) (ii) 3.348 - 2.104 (2 decimal places) (iii) 2.281 x 3.567 (2 decimal places).

(i) Approximated value: round the inputs to 1 decimal place first: 4.281 becomes 4.3 and 2.157 becomes 2.2.

Step: 4.3 / 2.2 = 1.9545..., which is 1.95 to 2 decimal places.

Step: Accurate value: 4.281 / 2.157 = 1.9846..., which is 1.98 to 2 decimal places.

Step: Difference = 1.98 - 1.95 = 0.03, which is small and of the same size as the numbers.

Final answer (i): approximated 1.95, accurate 1.98 — the solution IS reasonable.

(ii) Approximated value: round the inputs to 1 decimal place first: 3.348 becomes 3.3 and 2.104 becomes 2.1.

Step: 3.3 - 2.1 = 1.2.

Step: Accurate value: 3.348 - 2.104 = 1.244, which is 1.24 to 2 decimal places.

Step: Difference = 1.24 - 1.20 = 0.04, which is small.

Final answer (ii): approximated 1.2, accurate 1.24 — the solution IS reasonable.

(iii) Approximated value: round the inputs to 1 decimal place first: 2.281 becomes 2.3 and 3.567 becomes 3.6.

Step: 2.3 x 3.6 = 8.28.

Step: Accurate value: 2.281 x 3.567 = 8.136327..., which is 8.14 to 2 decimal places.

Step: Difference = 8.28 - 8.14 = 0.14, which is small compared with 8.14.

Final answer (iii): approximated 8.28, accurate 8.14 — the solution IS reasonable.